

Square $n \times n$ A

$$A \underline{x} = \underbrace{\lambda \underline{x}}_{\lambda I_n \underline{x}}$$

$$\frac{d e^{z x}}{d x} = z e^{z x}$$

$$A \underline{x} - \lambda I_n \underline{x} = \underline{0}$$

$$(A - \lambda I_n) \underline{x} = \underline{0}$$

Determinant of A $\det(A)$ $|A|$

$$\det: \mathbb{R}^{n \times n} \rightarrow \mathbb{R}.$$

$$1. \det \begin{pmatrix} \alpha \underline{x}^T + \beta \underline{y}^T \\ -\underline{r}_2^T \\ \vdots \\ -\underline{r}_n^T \end{pmatrix} = \det \begin{pmatrix} -\alpha \underline{x}^T \\ -\underline{r}_2^T \\ \vdots \\ -\underline{r}_n^T \end{pmatrix} + \det \begin{pmatrix} -\beta \underline{y}^T \\ -\underline{r}_2^T \\ \vdots \\ -\underline{r}_n^T \end{pmatrix}.$$

$$(\text{Linearity in a row}) = \alpha \det \begin{pmatrix} -\underline{x}^T \\ -\underline{r}_2^T \\ \vdots \\ -\underline{r}_n^T \end{pmatrix} + \beta \det \begin{pmatrix} -\underline{y}^T \\ -\underline{r}_2^T \\ \vdots \\ -\underline{r}_n^T \end{pmatrix}$$

$$2. \det \begin{pmatrix} \vdots \\ -\underline{r}_i^T \\ \vdots \\ -\underline{r}_j^T \\ \vdots \end{pmatrix} = - \det \begin{pmatrix} -\underline{r}_i^T \\ \vdots \\ -\underline{r}_j^T \\ \vdots \end{pmatrix} \quad (\text{swapping rows changes sign}).$$

$$\begin{pmatrix} -\underline{r}_1^T \\ -\underline{r}_2^T \\ \vdots \end{pmatrix} \xrightarrow{r_2 \leftarrow r_2 + \alpha r_1} \begin{pmatrix} -\underline{r}_1^T \\ -\underline{r}_2^T + \alpha \underline{r}_1^T \\ \vdots \end{pmatrix}$$

$$\det \begin{pmatrix} -\underline{r}_1^T \\ -\underline{r}_2^T + \alpha \underline{r}_1^T \\ \vdots \end{pmatrix} = \det \begin{pmatrix} -\underline{r}_1^T \\ -\underline{r}_2^T \\ \vdots \end{pmatrix} + \alpha \det \begin{pmatrix} -\underline{r}_1^T \\ \underline{r}_1^T \\ \vdots \end{pmatrix}$$

$$3. \det(I_n) = 1.$$

$$\det \begin{pmatrix} 0 \\ -\underline{r}_2^T \\ \vdots \\ -\underline{r}_n^T \end{pmatrix} = \det \begin{pmatrix} \underline{r}_2^T - \underline{r}_2^T \\ -\underline{r}_2^T \\ \vdots \\ -\underline{r}_n^T \end{pmatrix} = \det \begin{pmatrix} -\underline{r}_2^T \\ -\underline{r}_2^T \\ \vdots \\ -\underline{r}_n^T \end{pmatrix} - \det \begin{pmatrix} \underline{r}_2^T \\ -\underline{r}_2^T \\ \vdots \\ -\underline{r}_n^T \end{pmatrix} = 0.$$

Coro Navy:

If A has all pivot cols
If A has a free col

$\det(A) = \text{prod of pivots.}$

$$\det(A) = 0.$$

A is invertible iff $\det(A) \neq 0$.

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\text{row 1} = a \begin{pmatrix} 1 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \end{pmatrix}$$

$$\text{row 2} = c(1 \ 0) + d(0 \ 1)$$

$$\det(A) = \begin{pmatrix} a(10) + b(01) \\ c \quad d \end{pmatrix} = a \det \begin{pmatrix} 1 & 0 \\ c & d \end{pmatrix} + b \det \begin{pmatrix} 0 & 1 \\ c & d \end{pmatrix}.$$

$$a \det \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} + a \det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + b \det \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + b \det \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$$

$$\frac{ad(+1) + bc(-1)}{= ad - bc.}$$

$$\det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = a_{11} \begin{vmatrix} 1 & 0 & 0 \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + a_{12} \begin{vmatrix} 0 & 1 & 0 \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$+ a_{13} \begin{vmatrix} 1 & 0 & 0 \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$r_2 = a_{21} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + a_{22} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + a_{23} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$a_{11} \begin{vmatrix} 1 & 0 & 0 \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} a_{21} \begin{vmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + a_{11} a_{22} \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + a_{11} a_{23} \begin{vmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$= a_{11} a_{22} a_{33} \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} + a_{11} a_{23} a_{32} \begin{vmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix}$$

$$= a_{11} a_{22} a_{33} (1) + a_{11} a_{23} a_{32} (-1)$$

$$\begin{array}{ccc}
 1 & \underline{2} & 3 & +1 \\
 1 & \underline{3} & 2 & -1 \\
 2 & 1 & 3 & -1 \\
 2 & 3 & 1 & +1 \\
 3 & 1 & 2 & +1 \\
 3 & 2 & 1 & -1
 \end{array}$$

$$\begin{array}{l}
 123 \rightarrow 213 \rightarrow 231 \\
 123 \rightarrow 132 \rightarrow 231 \\
 123 \rightarrow 132 \rightarrow 312 \rightarrow 321 \rightarrow 231 \\
 \downarrow \\
 123 \rightarrow 132 \rightarrow 312 \\
 123 \rightarrow 321
 \end{array}$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$* \det(A) = \det(A^T)$$

$A: n \times n$.

$$= \sum_{\sigma} a_{1, \sigma(1)} a_{2, \sigma(2)} \dots a_{n, \sigma(n)} (-1)^{\text{sign}(\sigma)} \quad \sigma(1) \dots \sigma(n): \text{permutation of } 1 \dots n.$$

$$= \sum_{\tau} a_{\tau^{-1}(1), 1} a_{\tau^{-1}(2), 2} \dots a_{\tau^{-1}(n), n} (-1)^{\text{sign}(\tau^{-1})}$$

$$= \sum_{\tau} a_{\tau(1), 1} a_{\tau(2), 2} \dots a_{\tau(n), n} (-1)^{\text{sign}(\tau)} \quad (n! \text{ terms}).$$

$\therefore$ All properties on rows mentioned above also hold for columns.

$$* \det(AB) = \det(A) \det(B)$$

Consider $* f$ linear in each row.

$f: \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$. $* \text{ swap 2 rows, } f \text{ changes sign.}$

$* f(I) = 1$.

Fix B .

$$f(A) = \frac{\det(AB)}{\det(B)}$$

To show: $f(A)$

$$= \alpha f\left(\begin{pmatrix} x \\ -r_2 \\ \vdots \\ r_n \end{pmatrix}\right) + \beta f\left(\begin{pmatrix} y \\ -r_2 \\ \vdots \\ r_n \end{pmatrix}\right)$$

$$\text{Show. } \det\left[\left(\begin{pmatrix} \alpha x + \beta y \\ \vdots \end{pmatrix} B\right)\right] = \alpha \det\left[\left(\begin{pmatrix} x \\ \vdots \end{pmatrix} B\right)\right] + \beta \det\left[\left(\begin{pmatrix} y \\ \vdots \end{pmatrix} B\right)\right]$$

